

MUHAMMAD ASAD LAL

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ABOUT ME

I am currently pursuing a Master’s degree in Computer Science and Technology at Nanjing University of Science and Technology. I have experience in programming, research, and technical problem-solving, with a strong willingness to learn and adapt in a professional environment. I aim to contribute to cutting-edge technologies and have a positive impact on both organizational and societal settings.

ACADEMIC QUALIFICATION

- Master of Science in Computer Science & Technology

Nanjing University of Science and Technology, China

Research: “Optimization using Metaheuristic Algorithms”

Supervisor: Prof. Bo Huang

Sep 2024 - Apr 2027
- Bachelor of Science in Computer Science, CGPA 3.82/4.00 (Top in Class)

Government College University Faisalabad, Pakistan

Final Year Project: “Heart Disease Prediction (Android Application)”

Supervisor: Prof. Ali Raza Shah

Aug 2019 - Sep 2023

PUBLICATIONS

1. Muhammad Asad Lal, Muhammad Kashif, Bo Huang*. “Father-inspired metaheuristic: a hybrid framework with neighborhood repulsion and elite learning for continuous optimization.”

Evolutionary Intelligence, Vol. 19, Article 111, 2026.

DOI: 10.1007/s12065-026-01228-w
2. Muhammad Asad Lal, Muhammad Kashif, JianYong Lv, Bo Huang*. “Forman-Ricci-Curvature-Guided SHADE for Multi-Level Thresholding in Fundus Image Segmentation.”

Applied Soft Computing, 2026. Submitted.

RESEARCH WORK

- Optimization using Metaheuristic Algorithms

Nanjing University of Science and Technology, China

Sep 2024 - Present

Designed and implemented new metaheuristic algorithms in MATLAB and Python.

Proposed the Father-Inspired Metaheuristic algorithm for continuous optimization.

Developed Forman-Ricci-Curvature-Guided SHADE for multi-level thresholding in fundus image segmentation

Evaluated algorithm performance using benchmarks, convergence analysis, and quality metrics.

RESEARCH DIRECTIONS

- Optimization Algorithms

Image Processing

Pattern Recognition

Engineering Design Optimization

Machine Learning/Artificial Intelligence

Deep Learning Models

PROFESSIONAL EXPERIENCE

- Android Development, Internship

Idealogic Software House

Oct 2023 - Mar 2024

Contributed to Android mobile app development projects.

Worked with the team during the app development lifecycle.

Improved practical skills in Android development, programming, and software practices.

UNDERGRADUATE PROJECTS

1. **Heart Disease Prediction, Final Year Project** Aug 2019 - Sep 2023
Developed an AI-based Android app using Logistic Regression to classify users' heart health as "Healthy" or "Unhealthy" and provide basic recommendations.
2. **Deep Learning-based Face Recognition, Semester Project** Feb 2023 - May 2023
Developed a CNN-based face recognition model trained on image data to identify faces and patterns.

PROFESSIONAL AND PERSONAL SKILLS

- **Programming Languages:** Python, MATLAB, Java, and C++.
 - **Optimization & Research:** Metaheuristic algorithms, Continuous optimization, and Statistical analysis.
 - **Machine Learning & AI:** Machine learning models, data analysis, and AI-based problem solving.
 - **Image Processing:** Medical image segmentation, multi-level thresholding, and image quality evaluation.
 - **Android Development:** Android app development, API integration, Java, and Android Studio.
 - **Problem-solving & Adaptability:** Strong analytical thinking, Quick learning, and Research-oriented.
- Writing:** Academic writing, technical documentation, and research manuscript preparation.

AWARDS & CERTIFICATIONS

One Million Prompters, Certification <i>Dubai Future Foundation</i>	June 24, 2026
Chinese Writing Competition, Award Certificate <i>Nanjing University of Science and Technology</i>	Dec 31, 2024
Manage Sensitive Data and Security Policies within GitHub, Certification <i>Microsoft</i>	Apr 01, 2024
Outstanding Academic Performance Award <i>Government College University Faisalabad</i>	Sep 29, 2023
Machine Learning with Python, Certification <i>International Business Machines (IBM)</i>	Nov 22, 2023
Deep Learning with TensorFlow, Certification <i>International Business Machines (IBM)</i>	Nov 06, 2023
Computer Application and Office Professional, Certification <i>Government of Punjab Vocational Training Council</i>	Oct 17, 2017

EXTRA-CURRICULAR ACTIVITIES

Basic Community Safety Training <i>Punjab Emergency Service Rescue 1122, Pakistan</i>	Jun 15, 2023
BSAFE Security Awareness Training <i>United Nations Department of Safety and Security</i>	May 26, 2022

REFERENCE

Will be furnished on Request.